First Demonstration of Three-Dimensional Thermal Conductivity Distribution Measurements of Interconnect Stacks Down to 3nm Process Nodes

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***Abstract*—**The thermal properties of back-end-of-line (BEOL) interconnect stacks are increasingly important in the heat dissipation of advanced chips, particularly in three-dimensional integration circuits (3D-IC). In this work, for the first time, we experimentally determined the 3D thermal conductivity (TC) distribution of interconnect stacks via layer-by-layer time-domain thermoreflectance mapping (TDTR-mapping). Additionally, steady-state thermoreflectance (SSTR) was employed to measure the cumulative thermal resistance (R_{th}) across the layers. Surprisingly, we observed that the effective TCs of most interconnect layers for 6 chips with different process nodes (28 nm - 3 nm) are only 1-2 W/m-K, significantly lower than the TC of bulk copper (~400 W/m-K). Some values are even lower than the TC of SiO_2. Structural and theoretical analyses attribute this to the high thermal boundary resistance (TBR) at metal/dielectric interfaces, as well as dense grain boundaries and amorphous components within the metal wires. Thermal models were developed to predict the temperature distribution in double-side integrated transistors with a novel cooling strategy. This work provides a metrology for quantitatively evaluating thermal performance of device and packaging design.

I. Introduction

As the size of transistors continues to shrink, inefficient heat dissipation from the active layer (AL) has become a critical barrier for further increasing integration density [1]. Crucially, the growing complexity and thickness of BEOL interconnect stacks in 3D ICs have emerged as the primary thermal bottleneck [2-4]. Although copper wiring constitutes a significant portion of the interconnect stack—with a bulk TC of 400 W/m-K—the effective TC of the stack may be drastically reduced due to the presence of amorphous low-*k* dielectric and high density of interfaces. Despite its importance, the TC of interconnect layers remains poorly understood, lacking systematic study via advanced thermal characterization techniques.

In this work, TDTR and SSTR are performed to measure the TC and R_{th} of interconnect layers in various chips, including chips with process nodes of 28 and 14 nm, as well as 16 nm, 7 nm, 5 nm, and 3 nm. With a high-precision displacement stage, spatial TC distribution across interconnect layers is obtained. Grain boundaries and elemental distribution of copper wire were investigated by transmission electron microscope (TEM), which provided insights into the degradation mechanisms of the effective TC. Finally, finite element analysis (FEA) was applied to estimate the thermal performance of double-side device with new cooling strategies.

II. Experimental Setup and Sample Preparation

The experimental configurations of TDTR and SSTR are illustrated in **Figs. 2(a)** and **2(b)**, respectively. TDTR is an advanced technique for high-precision nanoscale thermal characterization [5-6]. In contrast, SSTR is effective in measuring the cumulative R_{th} across complex multilayer stacks [4]. **Fig. 3(a)** shows the sample preparation process. The original chips were delayered by Xe Plasma FIB with negligible damage to expose multiple interconnect layers. An ~80 nm Al transducer was then deposited onto the sample, and measurements were carried out sequentially from the bottom to the top layer. **Fig. 3(b)** shows the 7 nm chip sample after delayering. **Fig. 3(c)** shows the delayered region with surfaces of different interconnect layers, and **Fig. 3 (d)** shows the surface smoothness condition under CCD camera.

III. Results and Disscusion

The representative TDTR results of the 7 nm chip are shown in **Fig. 4(a)**, where the ratio signal is fitted using an analytical model to determine the TC of certain interconnect layers. To minimize coupling effects caused by the small thickness (≤100 nm), the TC of two adjacent layers was measured as a single effective value, including all Cu interconnects and their paired interlayer dielectric (ILD) layers. In the first layer, the TC of AL of some samples was incorporated into the thermal model, and the high Si proportion results in a relatively high effective TC. For the thick top layer (M8M9), measurements employed a lower modulation frequency for greater thermal penetration depth and more comprehensive thermal characterization. **Fig. 4(b)** displays a sensitivity analysis for M8M9, indicating high sensitivity to the M8M9 TC, while the influence of the underlying layers is negligible.

The same measurement procedure was applied for measuring the TC of interconnect stacks across all chips, as shown in **Fig. 5(a)-(f)**. For chips with relatively thick interconnect layers (>100 nm), the TC of each individual layer was measured separately. The results revealed that most interconnect layers exhibit remarkably low TC values (1-2 W/m-K). This was attributed to the reduced TC in nanoscale Cu wires and the high TBR at the copper/dielectric interfaces.

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Fig. 6 shows the elemental distribution in the interconnect stacks of 14 nm chip. The Ta elements at the Cu/dielectric interfaces were observed. These interfacial layers result in higher $TBR_{Cu/dielectric}$, which significantly reduces the effective TC of the stack. Complementary analysis in **Figs. 7(a)** and **7(b)** presents the structure of the metal wire for each layer. Small grain size, amorphous region, and large density of lattice defects significantly reduce the TC of copper wires. A thermal transport model based on Callaway mode and effective medium theory [7-8] was developed to quantify these effects. The TC_{Cu} of each layer of 7 nm and 14 nm chip was calculated based on the grain size estimated from TEM, with the grain boundary thermal resistance set to 0.1 m^2-K/GW [9]. $TBR_{Cu/dielectric}$ and $TC_{dielectric}$ of 7 nm and 14 nm chip are then obtained by fitting and the results are shown in **Fig. 8**. The fitted $TC_{dielectric}$ is in reasonable agreement with literature [10]. The fitted $TBR_{Cu/dielectric}$ is also consistent with reported values, as shown in **Table 1** [11-12], which indicates that the existence of interlayer significantly increases $TBR_{Cu/dielectric}$.

TDTR mapping of TC was performed on 7 nm and 3 nm chips. By optimizing the delay time to minimize Al/stack TBR sensitivity, the TDTR ratio directly reflected the TC of the layers. As shown in **Figs. 9(a)** and **9(b)**, this approach successfully resolved the three-dimensional TC distribution of interconnect stacks for the first time. The distinct thermal pattern of 7 nm M1 arises from its structure, which comprises materials with different TC values, as shown in **Figs. 10(a)** and **10(b).** The missing region of the map in 3 nm is due to the limited delayered area and surface defects. The mapping results show that the TC distribution is generally uniform except for some localized regions. The mapping step size and spatial resolution is ~10 μm. The 3 nm chip stack shows slightly reduced TC compared to 7 nm chip due to the higher interface density caused by thinner layers.

To estimate the thermal properties of the interconnect stacks under working conditions, the TC of the interconnect layers in 7 nm and 3 nm chips was measured at elevated temperatures, as shown in **Figs. 11(a)** and **11(b)**. The results show that the TCs of all layers do not vary significantly up to 120°C, since TBR and $TC_{dielectric}$ exhibit negligible temperature dependence within this temperature range [10, 12].

The cumulative R_{th} of various interconnect layers was measured by SSTR, a method based on the linear steady-state temperature response of the sample under various laser power. The fitting results is shown in **Fig. 12**, along with the calibration result based on SiO_2 (TC = 1.4 W/m-K). The cumulative R_{th} values of 7 nm and 3 nm chips are shown in **Fig. 13**. The R_{th} of 3 nm chip is slightly larger than 7 nm, which is consistent with TDTR. Good linearity was established, which corresponds to the similar TC across the interconnect layers. The equivalent oxide (SiO_2) thickness (EOT) of thermal resistance, namely Kaptiza length, reaches nearly 2 μm, highlighting the high R_{th} of the interconnect stacks.

IV. Finite Element Simulation

Based on the experimental results of 3 nm chip, FEA was performed by COMSOL to estimate the thermal management performance of a double-sided transistor configuration [13]. Three models incorporating measured interconnect stack properties were developed: a conventional single-side device structure with Cu heat sink integrated via thermal interface materials (TIM), and two double-sided device configurations with Cu/diamond heat sink and SiO_2/AlN dielectric bonding layer. Structure diagrams, mesh conditions, and representative temperature distributions are shown in **Fig. 14**, with component and material properties listed in **Table 2**.

Fig. 15 shows the peak temperature rise (T_{peak}) of the three models under various convection heat transfer coefficients (h_{con}), representing different cooling strategies. As h_{con} increases, thermal transport transitions from convection-dominated to conduction-limited regime, accompanied by a decreasing slope of the T_{peak} - h_{con} curve. In contrast, the direct integration of heat sink without TIM achieves substantially lower conduction R_{th}, which significantly alleviates the heat dissipation challenge in 3D ICs. particularly in high-TC dielectric, heat sink, and high h_{con} conditions. Under these optimized conditions, a more than 50% reduction in junction-to-ambient thermal resistance (θ_{JA}) can be achieved. The simulation results in **Fig. 16** further demonstrate that the novel diamond heat sink with AlN bonding layer strategy achieves a ~ 33% reduction in T_{peak} compared to low-TC bonding layer/heat sink case at the evaluated h_{con}, maintaining comparable thermal performance to conventional designs while enabling doubled computational power.

V. Conlusion

In summary, this study presents the first demonstration of three-dimensional TC distribution characterizations of FinFET BEOL interconnects across 28nm to 3nm nodes by layer-by-layer TDTR mapping. Uniform but remarkably low TC was observed for most layers in a series of chips with different process nodes. The high $TBR_{Cu/dielectric}$ results in this unexpectedly low effective TC for the stacks, posing significant thermal challenges for interconnect-intensive 3D-ICs. However, high-TC diamond heat sink and high TC bonding layer integration effectively mitigates this limitation. These findings offer a metrology for systematic thermal performance evaluation and optimization strategies for advanced logic devices, especially for 3D ICs.

VI. Acknowledgment

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References

[1] M. S. Lundstrom, *et al*., *Science*, vol. 378, p. 722, 2022. [2] R. Brain, *et al., IEDM*, p. 9.3.1, 2016. [3] W.-Y. Woon *et al.*, *IEDM*, p.1, 2023. [4] W.-Y. Woon *et al.*, *VLSI* p. 1, 2024. [5] D. G. Cahill, *Rev. Sci. Instrum.*, vol. 75, p. 5119, 2004. [6] Z. Cheng *et al.*, *Appl. Phys. Rev.*, vol. 11, p. 041324, 2024. [7] Z. M. Zhang, *Nano/Microscale Heat Transfer*, in Springer Int. Publ., 2020. [8] C.-W. Nan, *et al*., *J. Appl. Phys.* vol. 81, p. 6692, 1997. [9] Z. Cheng *et al.*, *ACS Appl. Mater. Interfaces*, vol. 10, p. 4808, 2018. [10] C. Stabler, *et al*., *Materials*, vol. 11, p. 279, 2018. [11] T. Z. Zhan *et al.*, *ACS Appl. Mater. Interfaces* vol. 12, p. 22347, 2020. [12] J. Hu *et al.*, *Int. J. Heat Mass Transf.*, vol. 251, p. 127385, 2025. [13] H. R. Lu, *et al.*, *VLSI* p. 1, 2024. [14] V. Sam, *et al.*, *Adv. Funct. Materials*, vol. 35, p. 2402662, 2025.

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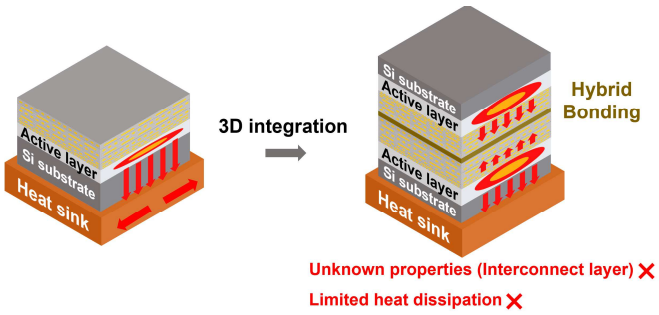


Fig. 1. BEOL interconnect stacks induced thermal dissipation challenge in 3D integration architecture

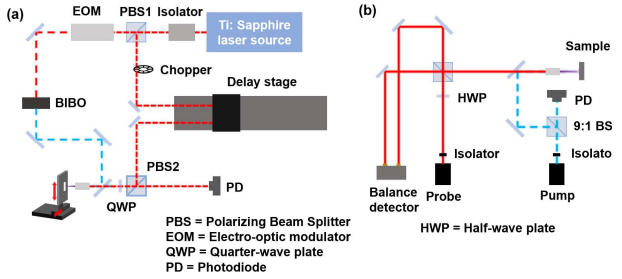


Fig. 2. The schematic diagrams of (a) TDTR and (b) SSTR

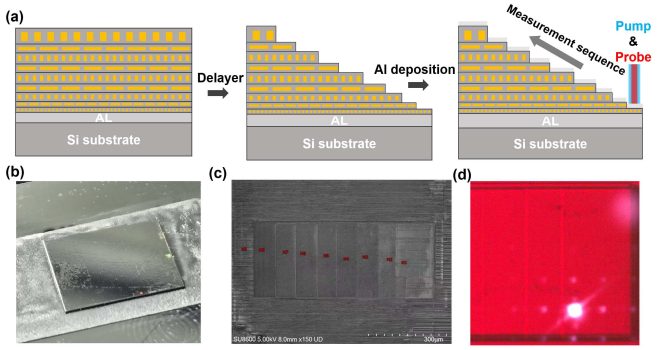


Fig. 3. (a) Sample preparation and measurement process (b) An appearance photo of 7 nm chip (c) The SEM image of the delayered area. (d) The image of the smooth surface in the CCD camera after coating Al.

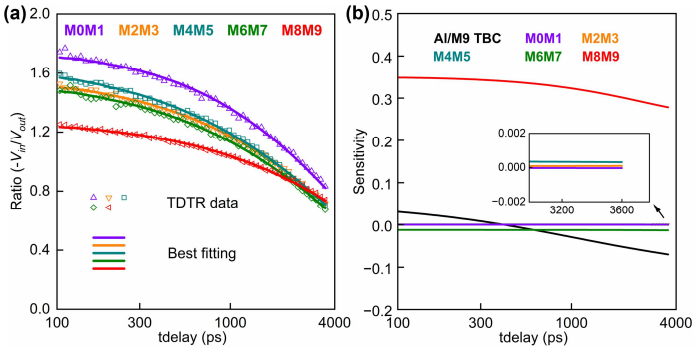


Fig. 4. (a) TDTR ratio data and best fitting result of 7 nm chip interconnect layers. (b) The sensitivities of different parameters in the measurement of M8M9. Thermal boundary conductance (TBC) is the reciprocal of TBR.

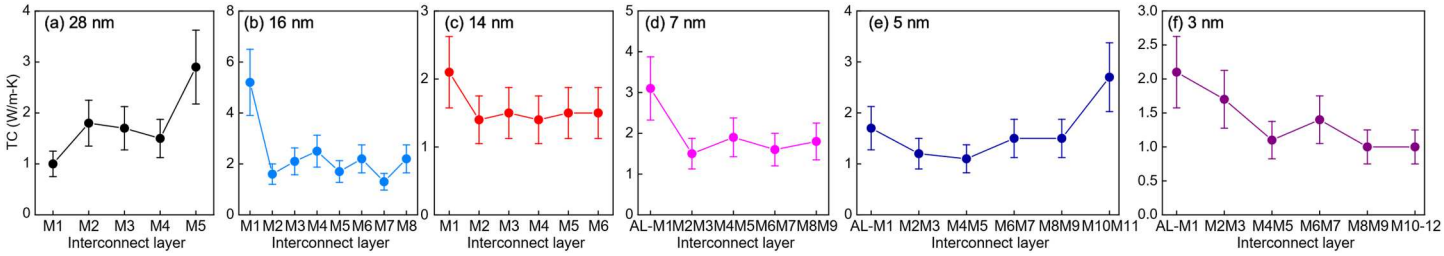


Fig. 5. TCs of different interconnect layers of (a) 28nm chip, (b) 16nm chip, (c) 14 nm chip, (d) 7 nm chip, (e)5 nm chip, (f) 3 nm chip.

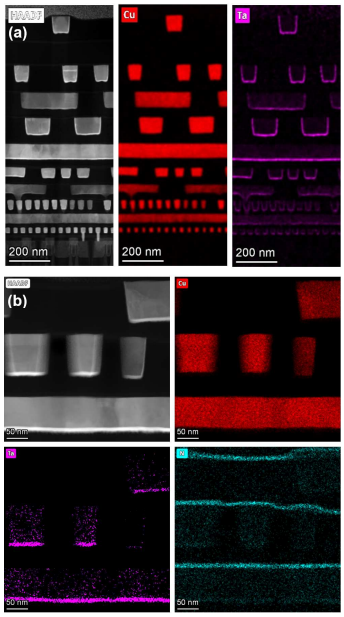


Fig. 6. Cu/dielectric interfacial elemental distribution of (a) 7 nm chip and (b) 14 nm chip.

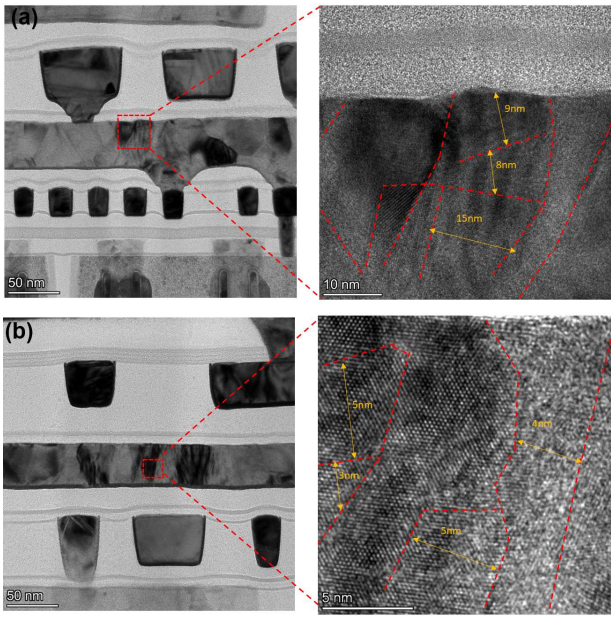


Fig. 7. Representative high-resolution TEM images of Cu grain size of (a) 7 nm M2 and (b) 7 nm M4

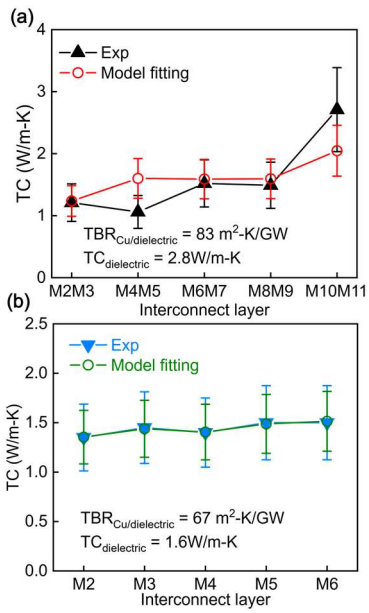


Fig. 8. Fitting results of the $TBR_{Cu/dielectric}$ and $TC_{dielectric}$ of (a) 7 nm and (b) 14 nm chip.

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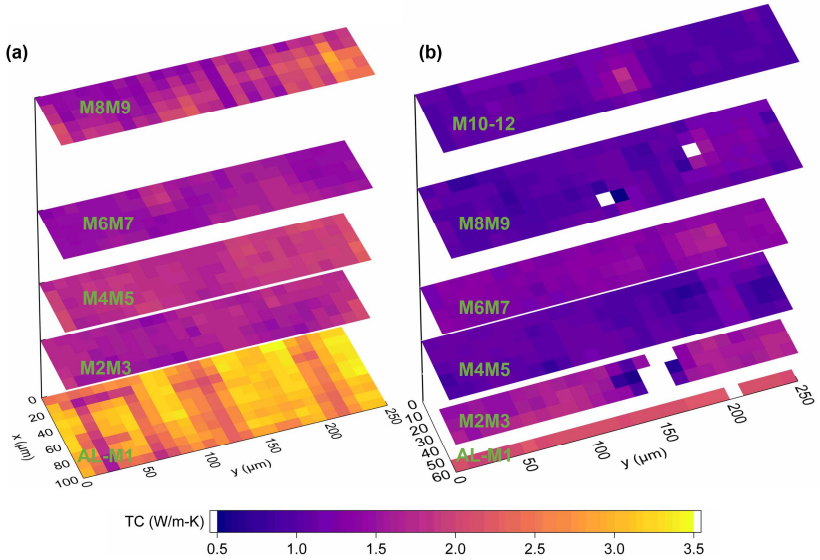


Fig. 9. Three-dimensional TC mapping of interconnect stacks (a) The TC distribution of 7 nm chip. (b) The TC distribution of 3 nm chip.

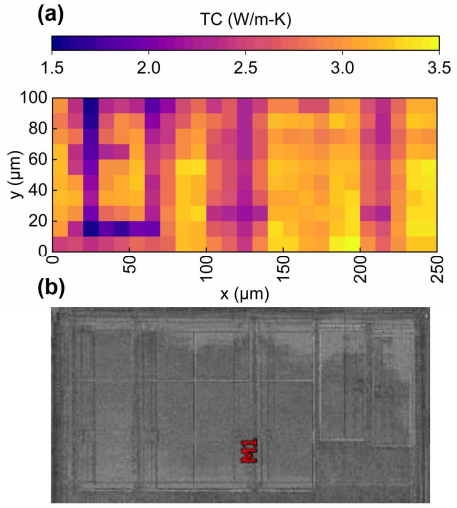


Fig. 10. (a) The thermal pattern of 7 nm M1 and (b) the corresponding circuit feature on M1.

Data source	Structure	TBR (m^2-K/GW)
This work	Cu/Ta/SiOC	83
This work	Cu/Ta/SiOC	67
Ref. [11]	Cu/Ta/SiO_2	99
Ref. [11]	Cu/TaN/SiO_2	62
Ref. [11]	Cu/TaN/Ta/SiO_2	64
Ref. [11]	Cu/Ta/TaN/Ta/SiO_2	29
Ref. [11]	Cu/Ta/TaN/SiO_2	17
Ref. [12]	Cu/SiO_2	9

Table 1. Summarized $TBRs_{Cu/dielectric}$ from pervious works and the corresponding interfacial structures.

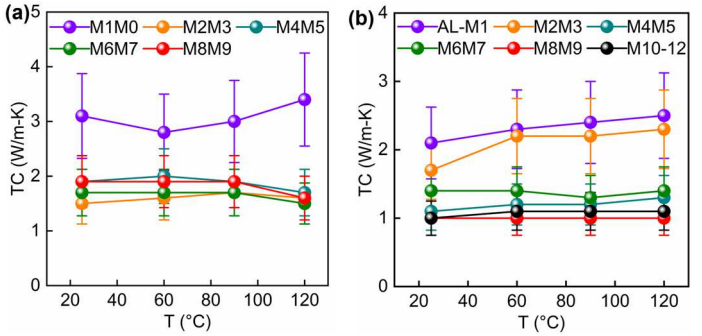


Fig. 11. The temperature dependence of effective TC of interconnect layers of (a) 7 nm chip and (b) 3 nm chip.

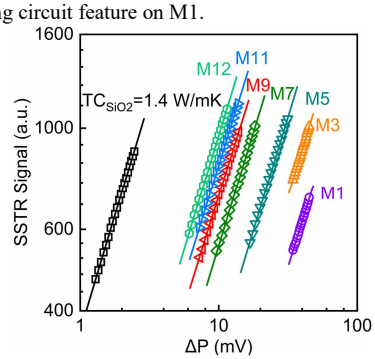


Fig. 12. The calibration and measurement results of different stack structure of 3 nm chip.

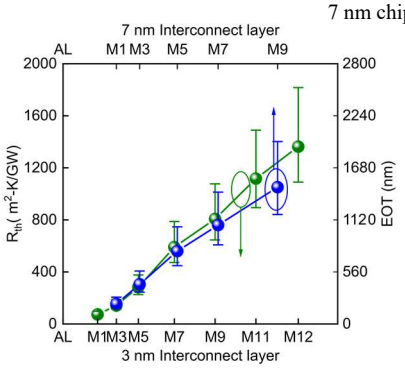


Fig. 13. The SSTR measured layer-cumulative R_{th} of 7 nm chip and 3 nm chip.

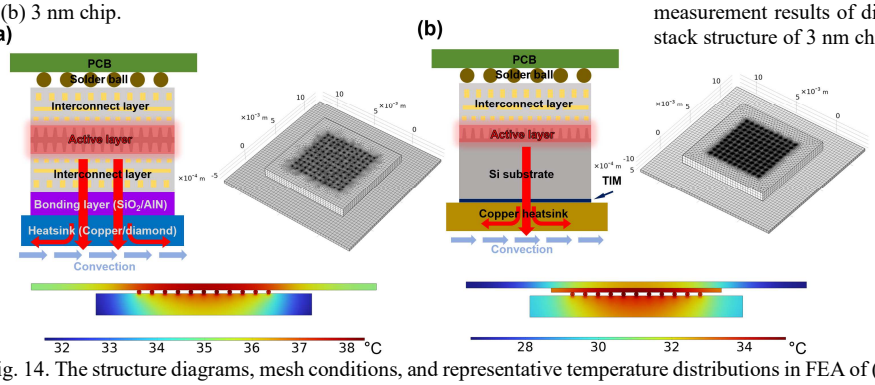


Fig. 14. The structure diagrams, mesh conditions, and representative temperature distributions in FEA of (a) Double-side device and (b) conventional package. More than one million mesh vertices were constructed in both cases.

Component	Material	TC (W/m-K)
Conventional substrate	Silicon	140
High-TC heat sink	Diamond	2200
Conventional heat sink	Cu	400
Bonding layer	AlN/SiO_2	90 (300nm)/1.4
Solder Ball	63Sn/37Pb	50
EMC	FR4	0.3
Substrate/Cu	R_{TIM}=15 m^2-K/MW	
Stack/AlN	TBR = 5 m^2-K/GW	
AlN/Diamond	TBR = 20 m^2-K/GW	
SiO_2/Cu	TBR = 20 m^2-K/GW	

Table 2. Summary of the important parameters of different components and structures in the FEA model. The TC of Cu and FR4 is taken from the database of COMSOL, TC_{AlN} and $TBR_{Stack/AlN}$ are taken from ref. [14].

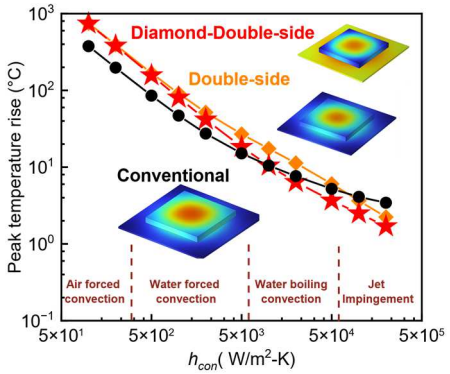


Fig. 15. Cooling performance of different h_{con} and strategies in double-side device and conventional package, the heat dissipation power of each AL is 10 W.

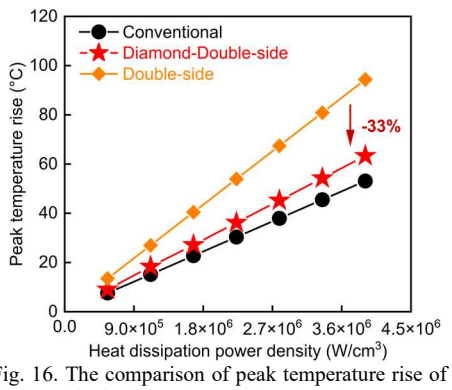


Fig. 16. The comparison of peak temperature rise of double-side device and conventional package under a h_{con} of 5000 W/m^2-K.

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